Liquidity, Price Dynamics, and Triangular Arbitrage

Evidence from NASDAQ and Foreign-Exchange Limit Order Books

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Abstract

This report studies one trading session for two NASDAQ stocks and the assigned EUR/USD–USD/JPY–EUR/JPY currency triangle. AAPL and GPRO were selected to expose the same measurement pipeline to two very different market structures: an actively traded large-cap stock and a thin, low-priced stock whose quoted spread is constrained by the one-cent tick. AAPL trades roughly \$4.42 million per minute, compared with only \$9.66 thousand for GPRO. The liquidity ranking is correspondingly sharp: mean five-second price impact is 0.56 basis points for AAPL and 13.25 basis points for GPRO. In foreign exchange, the main intraday event is the London close rather than the New York equity open. We also identify 28 one-second triangular-arbitrage observations grouped into 20 episodes. Their frictionless marked profit is economically small and remains an upper bound because the data do not include fees, last-look rejections, or execution latency. The follow-up instruction requests Monday–Friday liquidity plots; the present report is explicitly a one-day cross-sectional case study, and the weekly extension and its testable implications are stated below.

1 Introduction

The objective is to characterize trading activity, liquidity, short-horizon price dynamics, and executable triangular arbitrage using high-frequency order-book and transaction data. The equity sample contains AAPL and GPRO on 30 December 2019 from 09:30 to 16:00 Eastern Time. AAPL is a large-cap, high-turnover stock. GPRO is a small-cap stock trading near \$4.34, so its one-cent minimum tick is economically large. The comparison is therefore not a universe-wide stock-selection exercise. It is a controlled contrast between a thick, continuously refreshed book and a sparse, tick-constrained book processed with the same code.

The economic prior follows the information-asymmetry logic of Glosten and Milgrom [1] and Kyle [2]. When a dealer has less uninformed order flow against which to average informed trading, adverse-selection costs should appear in wider relative spreads and greater price impact. The AAPL–GPRO comparison is consistent with that prediction, but the intraday evidence requires more care: GPRO's spread is primarily a tick-size outcome, not a smooth information-driven U-shape.

The currency sample contains EUR/USD, USD/JPY, and EUR/JPY on 25 January 2012 over the same 09:30–16:00 clock. These pairs are prescribed by the assignment and form a closed conversion triangle. Because this interval overlaps both London and New York trading, foreign-exchange results should not be forced into a U.S.-equity opening narrative.

The main findings are:

- AAPL trades approximately 457 times more dollar volume per minute than GPRO. GPRO appears deeper in shares at the best bid, but AAPL is deeper in dollar terms.
- AAPL's spread and price impact are elevated at the open and then decline; its close does not recreate the second arm of a textbook U-shape on this holiday-week Monday.
- GPRO's one-cent spread binds nearly all day. Its 14:30 variance spike is a sparse-trading burst, not a simultaneous spread widening or depth collapse.
- The three currency pairs experience a synchronized spread and variance event around the London close. EUR/JPY is the widest and sparsest leg and exhibits the clearest one-minute reversal.
- Apparent triangular arbitrage is rare, brief, and small after accounting for realistic implementation limits.

2 Data and Measurement

2.1 Data construction

NASDAQ trades and ten-level market-by-price books are obtained from Databento. Market-by-order records supply new-order counts. Timestamps are converted from UTC to New York time and processed in exchange-event time. A ten-minute pre-open buffer seeds the 09:30 stock book. Currency books are reconstructed from the two supplied EBS files, using integer price ticks as book keys and treating each size update as a replacement of the quantity at that price.

Stock dollar volume is price times shares. Currency volume is converted to U.S. dollars: EUR/USD is already in dollar notional, USD/JPY size is in dollars, and EUR/JPY yen notional is divided by the same-second USD/JPY midpoint. Equity order counts include genuine displayed additions and exclude replacement-generated add messages. Hidden trades remain in transaction-based volume and price measures but cannot be counted as displayed new orders.

2.2 Scope relative to the five-day requirement

The assignment follow-up asks for the three liquidity measures—spread, depth, and price impact—to be plotted and compared over five consecutive trading days from Monday through Friday. The locked submission contains one NASDAQ session and one currency session. We therefore do not present the current figures as a completed weekly comparison. The narrower design deliberately emphasizes the cross section: how the same estimators behave for AAPL versus GPRO and across the three legs of the currency triangle, rather than how one ticker's intraday profile changes from day to day. This focus permits a detailed treatment of the tick constraint, sparse impact identification, dollar versus share depth, and market-specific event clocks, but it cannot identify persistent day-of-week patterns.

Expected results from a full week. A five-day extension would first plot each daily profile separately and then compare a cross-day mean profile with day-level dispersion bands. If the equity patterns are structural, AAPL should repeatedly open with wider spreads, higher impact, and greater variance; a less holiday-affected week may also restore the usual late-day rise that is absent on 30 December 2019. GPRO's one-cent spread should remain comparatively flat whenever the tick binds, while isolated variance bursts such as the 14:30 event should move across days or disappear if they are idiosyncratic. Repetition would distinguish a stable thin-name pattern from a single-day print. In foreign exchange, recurring spread and impact widening near the London close would support the event-clock interpretation; failure to recur would recast the 12:25 move as date-specific. Pooling five days would also provide more identified impact minutes for the sparse instruments, tighter uncertainty around intraday averages, and a more reliable estimate of triangular-arbitrage frequency and duration. These are predictions for the unobserved weekly panel, not results inferred from the one-day sample.

2.3 Liquidity measures

Let a_t and b_t denote the best ask and bid. The quoted spread is

$$s_t = a_t - b_t. \tag{1}$$

Because quotes are piecewise constant, spread and depth are averaged by the duration for which each book state is valid:

$$\bar{s}_\Delta = \frac{\sum_i s_i \,\Delta t_i}{\sum_i \Delta t_i}. \tag{2}$$

Figures report the spread in basis points of the corresponding five-minute VWAP,

$$s_\Delta^{\text{bps}} = 10^4 \frac{\bar{s}_\Delta}{\text{VWAP}_\Delta}. \tag{3}$$

Level-one depth is the displayed quantity at the best bid and ask, reported separately. Depth at twice the daily average spread sums all displayed bid quantities at prices above $m_t - 2\bar{s}_{\text{day}}$ and all ask quantities below $m_t + 2\bar{s}_{\text{day}}$, where $m_t = (a_t + b_t)/2$. Following the assignment convention, both band-depth measures are zero when the contemporaneous spread exceeds twice its daily time-weighted average. Stock band depth is limited to the ten observed levels; the diagnostics confirm that this limit does not truncate the band in the present sample.

Five-second price impact is estimated separately in each minute:

$$\frac{m_{t+5} - m_t}{m_t} = \alpha_\tau + \lambda_\tau \operatorname{sign}_t + \varepsilon_t, \qquad \operatorname{sign}_t \in \{-1, +1\}. \tag{4}$$

A minute is retained only when it contains at least five signed trades and both trade directions. This rule gives stable identification for AAPL and EUR/USD but leaves many missing estimates for GPRO and EUR/JPY. Reported impact means therefore refer only to identified minutes.

2.4 Prices, returns, and dependence

The last midpoint and transaction price in each one-second and one-minute interval are retained. After the first valid observation, the last price is carried forward through an empty interval. Log returns are

$$r_t = \log p_t - \log p_{t-1}, \tag{5}$$

and realized variance is $\sum_t r_t^2$. Autocorrelations are calculated through lag 10; lag 0, which equals one by construction, is omitted from the plot. Thirty-minute variance and lag-1 autocorrelation use the one-minute transaction-return series. An opening half-hour contains 29 returns; subsequent bins contain 30.

2.5 Triangular arbitrage

The one-second currency books are synchronized and required to be strictly positive and uncrossed. Two aggressive conversion loops are tested:

$$L_A = \frac{b_{\mathrm{EUR/USD}}\, b_{\mathrm{USD/JPY}}}{a_{\mathrm{EUR/JPY}}}, \tag{6}$$

$$L_B = \frac{b_{\mathrm{EUR/JPY}}}{a_{\mathrm{USD/JPY}}\, a_{\mathrm{EUR/USD}}}. \tag{7}$$

An opportunity exists when either loop exceeds one. Every leg crosses the spread: purchases execute at the ask and sales at the bid. Tradable quantity is the minimum BBO capacity across the three legs after conversion to euros. Marked profit is frictionless and therefore excludes fees, last look, communication delay, and partial fills.

3 Summary Statistics

Tables 1–4 report the number of available minute observations, mean, standard deviation, minimum, 25th percentile, median, 75th percentile, and maximum. Monetary and depth variables are scaled before display so that economically meaningful differences are visible without raw machine notation.

Table 1: Trading activity per minute

Instrument	Measure	N	Mean	Std. dev.	Min.	P25	Median	P75	Max.
Panel A: NASDAQ equities									
AAPL	Dollar volume ($ millions)	390	4.418	5.825	0.823	2.195	3.224	5.082	97.175
	Trades	390	171.99	123.69	50	101	135.5	193.8	1,272
	New orders	390	1,780.82	1,429.99	697	1,130.5	1,461.5	1,968.0	23,806
GPRO	Dollar volume ($ millions)	390	0.00966	0.02667	0	0	0.00043	0.00739	0.33232
	Trades	390	7.22	13.90	0	0	2	7.8	143
	New orders	390	68.30	135.83	2	17	32	87.8	2,171
Panel B: Foreign exchange									
EUR/USD	Dollar volume ($ millions)	390	105.613	144.871	0	35.013	64.851	113.909	1,818.583
	Trades	390	37.27	36.64	0	16	27	42	340
USD/JPY	Dollar volume ($ millions)	390	32.674	45.064	0	8.000	19.000	40.000	413.000
	Trades	390	15.49	16.85	0	5	11	20	155
EUR/JPY	Dollar volume ($ millions)	390	8.002	11.676	0	1.297	3.893	10.371	89.013
	Trades	390	3.95	4.95	0	1	2	5	37

Notes: Dollar volume is transaction price times executed quantity and is shown in millions of U.S. dollars per minute. Foreign-exchange orders are not reported because the assignment requests order counts only for NASDAQ ITCH. Zero-volume minutes are retained.

The activity contrast motivates the stock selection. AAPL averages $4.42 million and 172 trades per minute, while GPRO averages only $9.66 thousand and seven trades. GPRO's median minute contains two trades, and its 25th percentile is zero; AAPL's least active minute still contains 50 trades. EUR/USD is the most active currency pair, followed by USD/JPY and then EUR/JPY.

Table 2: Minute-level transaction prices

Instrument	Price measure	N	Mean	Std. dev.	Min.	P25	Median	P75	Max.
Panel A: NASDAQ equities (U.S. dollars per share)									
AAPL	Open	390	290.2103	2.0734	285.4100	288.3525	291.3350	291.6950	292.6000
	Close	390	290.2137	2.0746	285.3400	288.3563	291.3450	291.6975	292.6100
	High	390	290.3270	2.0413	285.5400	288.5425	291.4350	291.7800	292.6900
	Low	390	290.0994	2.1110	285.2300	288.2375	291.2600	291.6200	292.5500
	VWAP	390	290.2168	2.0760	285.4058	288.4042	291.3432	291.7045	292.6158
GPRO	Open	285	4.3359	0.0689	4.2000	4.2850	4.3200	4.3900	4.4650
	Close	285	4.3356	0.0693	4.2100	4.2850	4.3200	4.3900	4.4700
	High	285	4.3377	0.0685	4.2100	4.2900	4.3200	4.3900	4.4700
	Low	285	4.3336	0.0697	4.2000	4.2850	4.3150	4.3900	4.4650
	VWAP	285	4.3356	0.0690	4.2100	4.2850	4.3200	4.3900	4.4699
Panel B: Foreign exchange (quote currency per unit of base currency)									
EUR/USD	Open	389	1.302706	0.005592	1.294880	1.297390	1.303290	1.308100	1.312090
	Close	389	1.302735	0.005604	1.294900	1.297400	1.303400	1.308120	1.312090
	High	389	1.302942	0.005630	1.295290	1.297500	1.303660	1.308400	1.312090
	Low	389	1.302530	0.005571	1.294700	1.297200	1.302920	1.307920	1.311800
	VWAP	389	1.302739	0.005602	1.294931	1.297339	1.303268	1.308174	1.311956
USD/JPY	Open	377	77.9485	0.2226	77.5700	77.7300	77.8960	78.1730	78.2820
	Close	377	77.9471	0.2230	77.5700	77.7300	77.8920	78.1740	78.2870
	High	377	77.9571	0.2202	77.5990	77.7490	77.9070	78.1800	78.2880
	Low	377	77.9389	0.2254	77.5600	77.7280	77.8710	78.1700	78.2760
	VWAP	377	77.9479	0.2227	77.5808	77.7340	77.8877	78.1725	78.2810
EUR/JPY	Open	307	101.5338	0.1825	101.2000	101.4000	101.4960	101.6755	101.9400
	Close	307	101.5348	0.1831	101.2030	101.3985	101.5000	101.6800	101.9400
	High	307	101.5424	0.1828	101.2030	101.4005	101.5000	101.6900	101.9490
	Low	307	101.5267	0.1835	101.2000	101.3900	101.4930	101.6720	101.9400
	VWAP	307	101.5350	0.1829	101.2001	101.4005	101.5000	101.6791	101.9400

Notes: Statistics summarize one-minute transaction-based open, close, high, low, and volume-weighted average prices. $N < 390$ denotes minutes without a transaction; those observations remain missing rather than being imputed in this table.

Table 3: NASDAQ liquidity statistics

Instrument	Measure	N	Mean	Std. dev.	Min.	P25	Median	P75	Max.
AAPL	BBO spread (cents)	390	2.6357	0.5805	1.4646	2.2523	2.5712	2.8756	5.3829
	Bid depth (000 shares)	390	0.2482	0.1367	0.1109	0.1867	0.2165	0.2609	1.5324
	Ask depth (000 shares)	390	0.2229	0.1157	0.1118	0.1669	0.1970	0.2413	1.6478
	Bid depth, $2\bar{s}$ (000 shares)	390	1.2890	0.5334	0.4061	1.0022	1.1876	1.4375	5.7259
	Ask depth, $2\bar{s}$ (000 shares)	390	1.3445	0.7385	0.3857	0.9836	1.1803	1.4761	7.5540
	Five-second impact (bps)	390	0.5588	0.3640	-0.6182	0.3085	0.5306	0.7481	2.8159
GPRO	BBO spread (cents)	390	1.0049	0.0387	1.0000	1.0000	1.0000	1.0000	1.5272
	Bid depth (000 shares)	390	6.2724	5.3185	0.1971	3.3175	4.7893	7.2681	48.3086
	Ask depth (000 shares)	390	7.4832	5.3576	0.4980	3.6800	5.9441	9.5724	27.2236
	Bid depth, $2\bar{s}$ (000 shares)	390	14.7122	7.4023	1.8160	9.8009	13.1622	17.9248	59.1186
	Ask depth, $2\bar{s}$ (000 shares)	390	14.4058	7.0990	1.7790	9.0795	12.9261	18.6620	40.0157
	Five-second impact (bps)	51	13.2520	10.3977	-1.5549	5.7039	11.5607	20.2828	42.2978

Notes: BBO values are from the NASDAQ TotalView book, not the protected NBBO. Spread and depth are exact event-time-weighted minute means. The $2\bar{s}$ rows sum displayed quantity inside twice the daily average-spread band and are set to zero when the contemporaneous spread exceeds $2\bar{s}$. Impact is $10^4\widehat{\lambda}_\tau$ and is available only in minutes meeting the signed-trade identification rule.

Table 4: Foreign-exchange liquidity statistics

Instrument	Measure	N	Mean	Std. dev.	Min.	P25	Median	P75	Max.
EUR/USD	BBO spread (pips)	390	1.1334	0.1674	0.6706	1.0164	1.1235	1.2405	1.6425
	Bid depth (millions base)	390	2.0304	1.0136	1.0067	1.5238	1.7871	2.2530	10.5385
	Ask depth (millions base)	390	1.9764	0.8159	1.0400	1.5204	1.8240	2.2074	10.0017
	Bid depth, $2\bar{s}$ (millions base)	390	22.2909	7.4099	9.6717	17.9954	21.9975	26.1454	94.4548
	Ask depth, $2\bar{s}$ (millions base)	390	22.1599	14.2849	9.8717	16.9296	20.2950	24.0926	239.5783
	Five-second impact (bps)	386	0.1964	0.1635	-0.4536	0.0978	0.1780	0.2747	1.0021
USD/JPY	BBO spread (pips)	390	0.8007	0.1580	0.4027	0.6884	0.7947	0.8890	1.4418
	Bid depth (millions base)	390	2.0955	1.0207	1.0000	1.4542	1.8133	2.4013	7.5017
	Ask depth (millions base)	390	2.2145	1.5091	1.0000	1.4542	1.8375	2.3129	15.2817
	Bid depth, $2\bar{s}$ (millions base)	390	20.1592	6.1154	8.9000	15.1688	19.6667	24.7271	44.7550
	Ask depth, $2\bar{s}$ (millions base)	390	22.2068	12.7433	6.0667	14.5136	20.3136	24.7524	100.2245
	Five-second impact (bps)	302	0.2248	0.3061	-0.4050	0.0841	0.1912	0.3165	4.3045
EUR/JPY	BBO spread (pips)	390	1.8982	0.3301	0.7552	1.7035	1.9172	2.0892	3.3197
	Bid depth (millions base)	390	1.4653	0.5523	1.0000	1.1408	1.3067	1.5833	6.7900
	Ask depth (millions base)	390	1.9656	3.5727	1.0000	1.1642	1.3683	1.7071	48.9283
	Bid depth, $2\bar{s}$ (millions base)	390	16.9464	2.8538	8.0167	14.8579	16.8817	18.6617	30.7417
	Ask depth, $2\bar{s}$ (millions base)	390	18.9312	8.9422	10.2917	14.2763	16.9467	20.6650	90.1317
	Five-second impact (bps)	103	0.4630	0.4342	-0.3606	0.1730	0.4167	0.6614	1.9398

Notes: One pip is 10^{-4} for EUR/USD and 10^{-2} for the JPY pairs. Depth is in millions of base currency. EUR/USD, USD/JPY, and EUR/JPY have mean relative spreads of approximately 0.87, 1.03, and 1.87 basis points, respectively. Extreme ask-depth maxima are short-lived resting orders rather than typical book states.

The stock liquidity table reverses the ranking suggested by raw shares. GPRO displays about 6,272 shares at the bid, versus 248 for AAPL, but multiplying by price gives approximately \$27 thousand for GPRO and \$72 thousand for AAPL. AAPL's $2\bar{s}$ bid depth is 5.19 times L1 and its ask depth is 6.03 times L1. GPRO's corresponding ratios are only 2.35 and 1.93 because its average spread is already the minimum tick. EUR/USD lies at the other extreme: roughly 11 times as much displayed size sits inside the $2\bar{s}$ band as at the touch.

Spread and depth comove negatively in every instrument. The L1 correlations range from −0.10 for USD/JPY to −0.32 for EUR/JPY, while the nearby-book correlations are more negative, reaching −0.47 for EUR/JPY. Thus, a wider quote tends to coincide with less displayed quantity nearby, although the relation is not deterministic at the best price alone.

4 Intraday Liquidity

4.1 NASDAQ equities

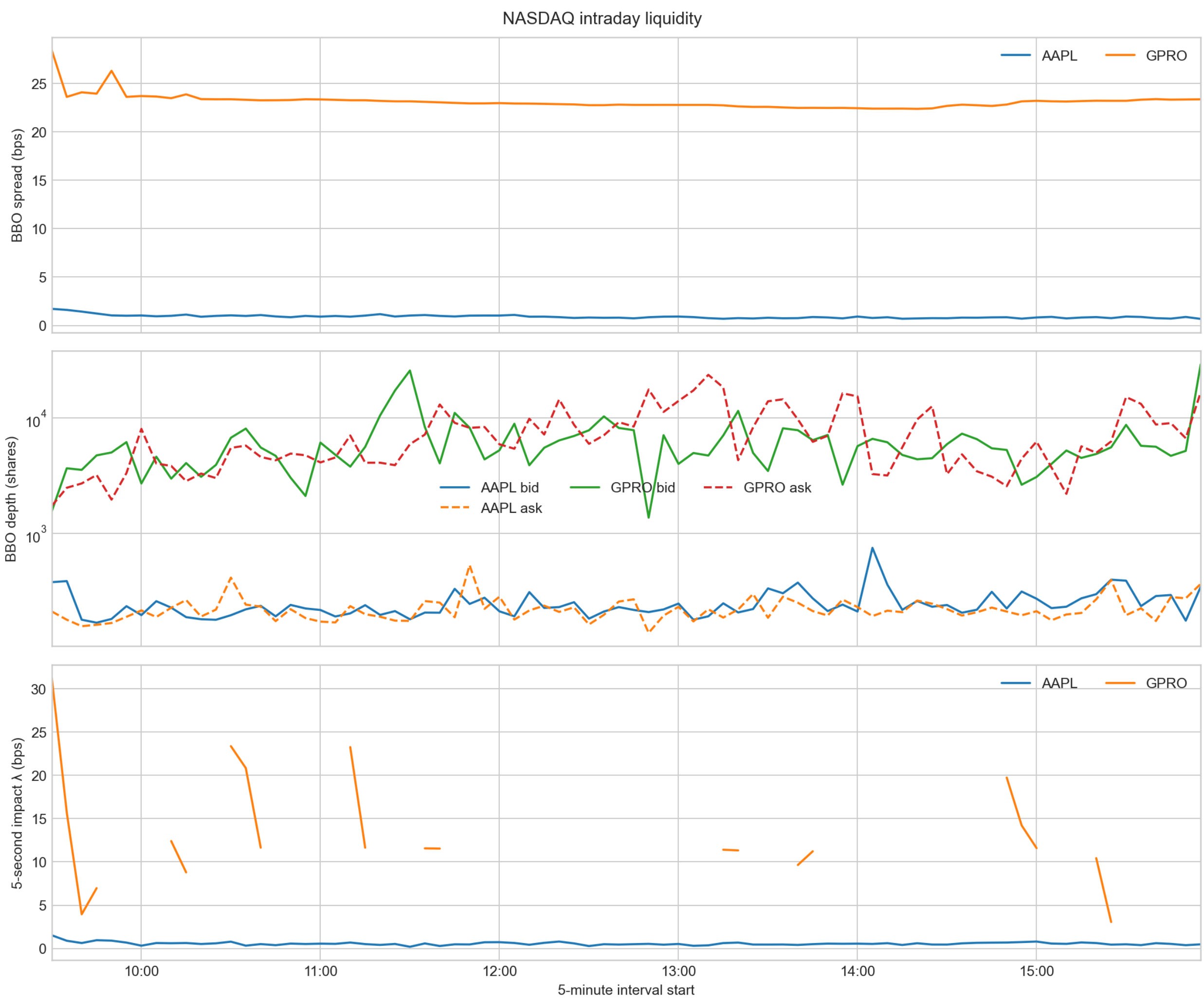

Figure 1: NASDAQ five-minute liquidity. Spread and bid/ask depth are exact event-time-weighted means; impact is the mean of available minute-level signed-impact slopes. Spread is in basis points of five-minute VWAP.

Spread. Figure 1 shows that AAPL begins the session with a visibly wider spread, including a sharp disturbance around 09:45, and then settles toward roughly 0.8–1.0 basis points. Its mean opening-half-hour spread is \$0.0385, compared with \$0.0253 during the rest of the session. The late-day spread does not rebuild a second arm of a U-shape. On 30 December 2019, the Monday of a short New Year week, this is a day-specific holiday-week pattern rather than evidence against the usual intraday regularity.

GPRO behaves differently. Its spread is one cent at the 25th percentile, median, and 75th percentile, with a standard deviation of only 0.039 cents. On a \$4.34 stock, one cent is approximately 23 basis points. The nearly flat line is therefore not evidence of invariant information risk; it is evidence that the tick prevents further tightening.

Depth. Bid and ask depth cross repeatedly because the displayed imbalance changes sign. For AAPL, the touch is a small, continuously refreshed slice of a deeper multi-level book. GPRO displays more shares but less dollar value. This distinction explains why raw share depth should not be interpreted as greater economic liquidity. The $2\bar{s}$ statistics reinforce the point: the nearby book adds five to six times L1 for AAPL but only about twice L1 for GPRO.

Price impact. AAPL has an impact estimate in every minute and averages 0.56 basis points. Impact is 0.93 basis points during the first 30 minutes and 0.53 thereafter, consistent with faster information revelation and more expensive immediacy at the open. GPRO averages 13.25 basis points, but this mean is based on only 51 identified minutes. The median GPRO minute contains two trades, 105 minutes contain no trade, and a one-tick midquote movement is already about 23 basis points. Consequently, the disconnected GPRO line reflects both missing regressions and economically coarse, sparsely identified price moves. It should not be read as a continuously observed high-frequency signal.

4.2 Foreign exchange

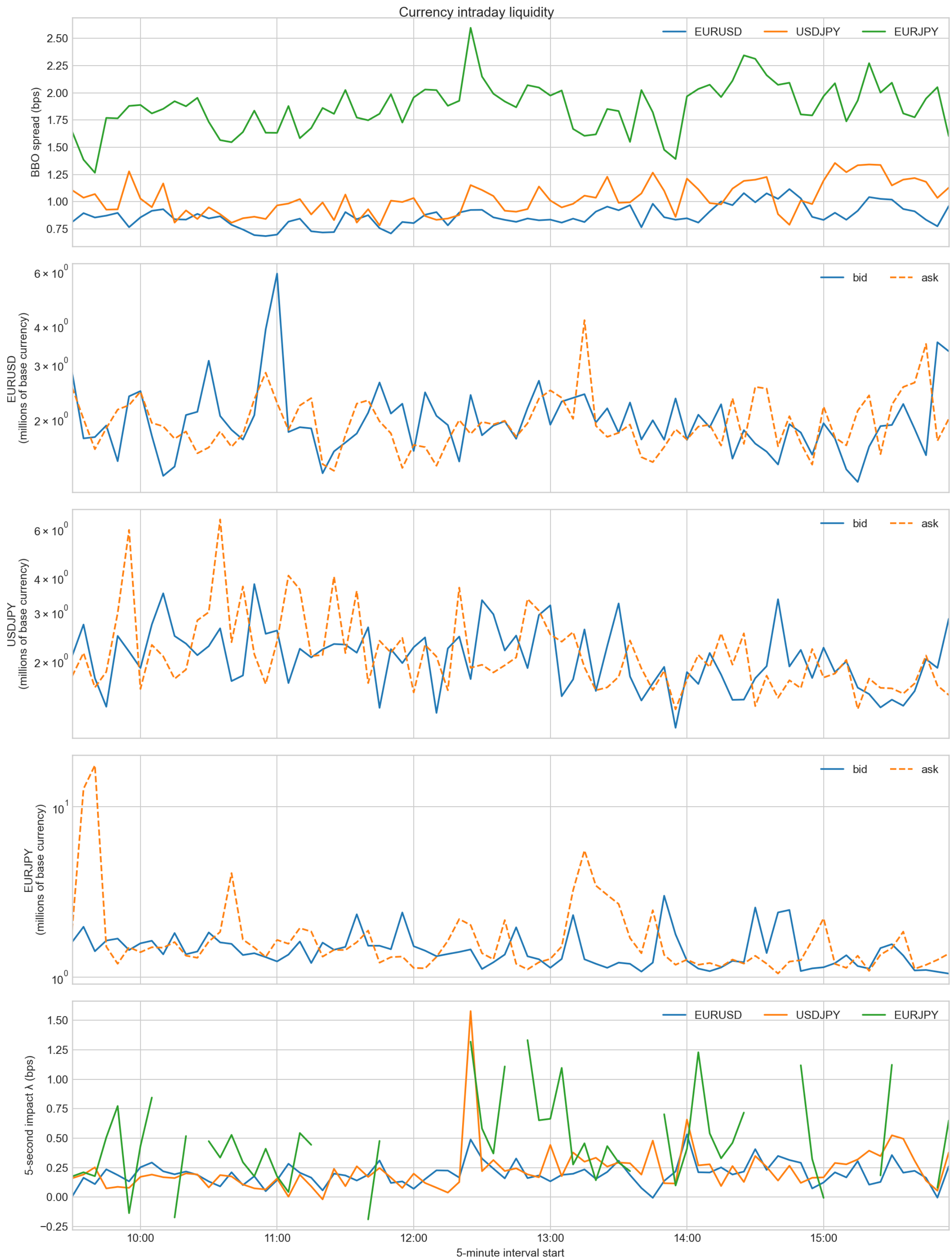

Figure 2: Foreign-exchange five-minute liquidity. Depth is split by pair to keep bid/ask movements legible. The session clock is 09:30–16:00 on 25 January 2012.

Spread. The spread ranking is EUR/USD, USD/JPY, then EUR/JPY: approximately 0.87, 1.03, and 1.87 basis points. The cross is the synthetic residual of the two dollar legs and has the wider, sparser standalone book. All three remain far tighter than GPRO, while EUR/USD is comparable to AAPL in relative terms. The synchronized widening around 12:25—EUR/JPY reaches roughly 2.6 basis points—points to a common London-close event rather than three independent liquidity holes.

Depth. EUR/USD and USD/JPY generally display one to four million units at the touch. EUR/JPY is thinner, with median bid and ask depth near 1.3–1.4 million. The large EUR/JPY ask observation before 10:00 reaches nearly 49 million, but its 75th percentile is only 1.71 million; it is a short-lived posted order, not the typical state of the book. Roughly ten to eleven times L1 depth lies inside the $2\bar{s}$ band for all three pairs.

Price impact. EUR/USD is identified in 386 of 390 minutes and remains near 0.20 basis points. USD/JPY is identified in 302 minutes and experiences a pronounced impact spike at the same 12:25 time as the spread event. EUR/JPY resembles GPRO in data coverage, not in magnitude: it has 83 zero-trade minutes and only 103 impact estimates. Its mean identified impact is 0.46 basis points, more than twice EUR/USD. Impact is lower in the first 30 minutes than afterward for every currency pair, which is the opposite of AAPL and confirms that 09:30 New York is not the organizing event for this FX sample.

5 Return Variation and Autocorrelation

Table 5: Return diagnostics by sampling frequency

Instrument	Frequency	Series	N	Mean	Std. dev.	Realized variance	$\rho(1)$
Panel A: One-second log returns							
AAPL	1 second	Midquote	23,399	3.302e−7	6.988e−5	1.143e−4	0.063
		Transaction	23,399	3.354e−7	7.348e−5	1.264e−4	0.037
GPRO	1 second	Midquote	23,399	5.001e−8	2.337e−4	1.278e−3	0.015
		Transaction	23,399	1.502e−7	2.103e−4	1.035e−3	0.072
EUR/USD	1 second	Midquote	23,399	4.602e−7	3.166e−5	2.345e−5	0.020
		Transaction	23,393	4.556e−7	3.537e−5	2.927e−5	-0.021
USD/JPY	1 second	Midquote	23,399	−1.764e−7	2.542e−5	1.512e−5	0.035
		Transaction	23,391	−1.773e−7	3.001e−5	2.106e−5	-0.013
EUR/JPY	1 second	Midquote	23,399	2.867e−7	3.077e−5	2.216e−5	0.046
		Transaction	23,340	2.855e−7	3.171e−5	2.347e−5	0.015
Panel B: One-minute log returns							
AAPL	1 minute	Midquote	389	2.484e−5	5.715e−4	1.270e−4	0.105
		Transaction	389	2.497e−5	5.823e−4	1.318e−4	0.078
GPRO	1 minute	Midquote	389	−1.200e−5	1.880e−3	1.371e−3	0.015
		Transaction	389	−9.004e−6	1.802e−3	1.259e−3	0.044
EUR/USD	1 minute	Midquote	389	2.809e−5	2.712e−4	2.885e−5	0.011
		Transaction	389	2.799e−5	2.748e−4	2.960e−5	-0.005
USD/JPY	1 minute	Midquote	389	−1.059e−5	2.695e−4	2.823e−5	0.004
		Transaction	389	−1.076e−5	2.739e−4	2.916e−5	-0.003
EUR/JPY	1 minute	Midquote	389	1.743e−5	2.425e−4	2.294e−5	-0.177
		Transaction	389	1.713e−5	2.481e−4	2.400e−5	-0.149

Notes: Returns are log differences of last-in-interval prices, with previous-tick carry-forward after the first valid observation. Realized variance is the unannualized sum of squared intraday returns. $\rho(1)$ is the full-session lag-1 sample autocorrelation.

Mean returns are small relative to their standard deviations. GPRO has substantially greater realized variance than AAPL at both horizons. Transaction-return variance is slightly above midquote variance for AAPL and the two dollar FX legs, consistent with bid–ask bounce. GPRO is the one-second exception: quote updates continue during seconds with no trade, so midquote realized variance exceeds transaction realized variance.

AAPL's one-minute midquote autocorrelation of 0.105 indicates mild continuation rather than reversal. Its five-second impact is therefore not obviously undone at the next minute. GPRO's corresponding coefficient is only 0.015, and the limited impact coverage prevents a strong permanence claim. EUR/JPY has one-minute transaction and midquote coefficients of −0.149 and −0.177, respectively, indicating temporary reversal on the thinner cross. EUR/USD and USD/JPY remain near zero, so this is not a general foreign-exchange finding.

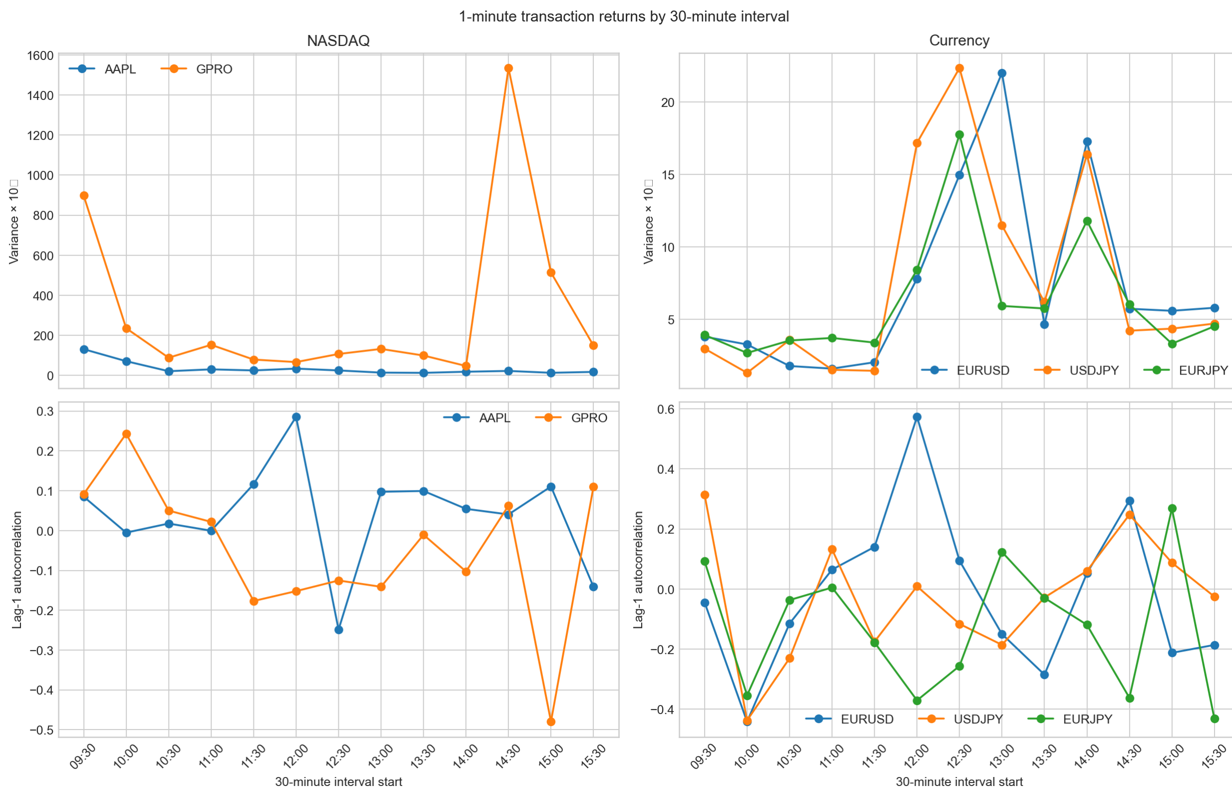

Figure 3: Thirty-minute variance and lag-1 autocorrelation of one-minute transaction log returns.

AAPL versus GPRO variance. AAPL's opening-half-hour variance is 1.31e−6; it then declines by approximately one order of magnitude and remains low. Lunch is near 1.3e−7, and the final half-hour is 1.79e−7. This path aligns with the wider opening spread and higher opening impact in Figure 1, but not with a closing liquidity deterioration.

GPRO opens at 8.98e−6, already about seven times AAPL's opening variance. Its largest observation is instead 14:30–15:00: 1.53e−5, 32 times the preceding half-hour and about 67 times AAPL at the same clock time. The spread remains at one cent and bid depth rises rather than collapses, while the impact regression is unidentified around the event. The coherent interpretation is a sparse, one-sided transaction burst whose few large prints dominate a half-hour containing many carried-forward zero returns. The next bin remains volatile and has autocorrelation −0.48, indicating that much of the move is reversed rather than permanently incorporated.

Currency variance. All three currency pairs peak between 12:00 and 13:30. USD/JPY reaches 2.23e−7 at 12:30, EUR/USD reaches 2.20e−7 at 13:00, and EUR/JPY reaches 1.78e−7 at 12:30. This is the same interval as the common spread widening and USD/JPY impact spike. Depth does not collapse at the same time, so variance comoves more clearly with spread and impact than with L1 depth.

Thirty-minute autocorrelation. Each coefficient uses only 29 or 30 returns, giving an approximate standard error of $1/\sqrt{30} \simeq 0.18$. Individual dots are therefore noisy. AAPL remains small throughout. GPRO's −0.48 coefficient at 15:00 is notable because it follows the 14:30 variance burst. Several EUR/JPY bins are materially negative, consistent with the full-session reversal in Table 5. The isolated EUR/USD coefficient of 0.57 at 12:00 is better read as temporary continuation into the London-close event than as stable predictability.

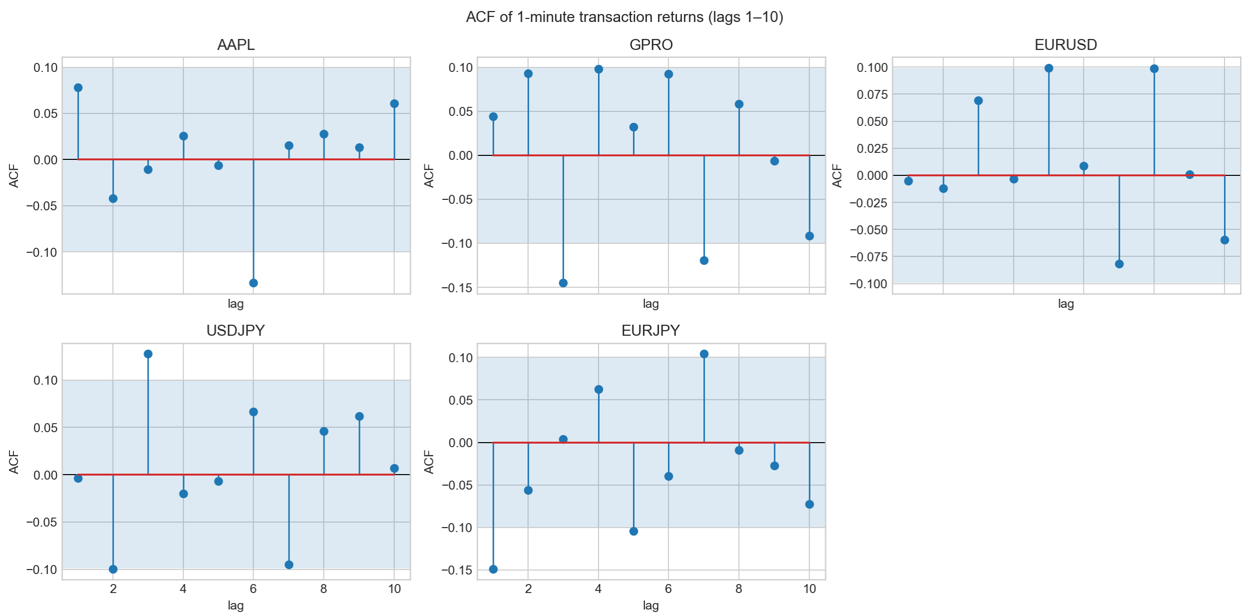

Figure 4: Autocorrelation of one-minute transaction log returns, lags 1–10. The shaded band is $\pm 1.96/\sqrt{T}$. Lag 0 is omitted.

At lag 1, AAPL and GPRO are mildly positive but remain inside the full-sample noise band; EUR/USD and USD/JPY are effectively zero. EUR/JPY is the only lag-1 coefficient outside the band, at −0.149. Most dependence disappears by lags 2–3. The occasional later stem outside the band is isolated rather than persistent, so the ten-lag plot provides little evidence of a stable forecasting rule.

6 Executable Triangular Arbitrage

Table 6: One-second triangular-arbitrage summary

Statistic	Value
Eligible seconds	28
Fraction of 09:30–16:00 session	0.120%
Distinct episodes	20
Mean episode duration	1.4 seconds
Maximum episode duration	6 seconds
Mean edge	0.187 bps
Maximum edge	1.181 bps
Mean binding quantity	1.273 million EUR
Maximum binding quantity	3.000 million EUR
Frictionless marked profit, all eligible seconds	$678.14

Notes: Every leg executes aggressively at the observed BBO. Twenty-five eligible seconds use Loop A and three use Loop B. Profit assumes full simultaneous execution at the displayed size and excludes all trading frictions.

6.1 A six-second episode

The longest episode begins at 09:36:35. In each of six consecutive seconds, Loop A sells euros for dollars at the EUR/USD bid, sells the dollars for yen at the USD/JPY bid, and buys the euros back at the EUR/JPY ask. The loop product is 1.00000397037, or 0.040 basis points. Binding size is EUR 2 million for the first three seconds and EUR 3 million for the last three. Full execution every second would mark approximately $77 before costs. Loop B remains below one throughout, so the result is one-sided rather than a crossed-book artifact.

The economically largest observation occurs at 09:43:32. Loop A widens from 0.18 to 0.49 and then 1.18 basis points over three seconds, with a binding size near EUR 771 thousand. The peak second marks about $118 and disappears one second later.

Across the day, the 28 eligible seconds cluster in two windows, 09:34–09:43 and 13:15–14:18. The afternoon cluster overlaps the tail of the London-close variance event and its 14:00 aftershock, but the clocks are not identical. Almost all observations use Loop A.

6.2 Economic feasibility

The $678 total is an upper bound, not cash that was necessarily available. The six-second episode is persistent but only 0.04 basis points wide, equivalent to $10–$15 per second at the displayed quantity. A last-look rejection, one-second delay, brokerage charge, or partial fill can remove the gain. The 1.18-basis-point observation is more attractive but lasts only one second. Since the files do not reveal exchange-to-user latency, firm-quote status, fees, or simultaneous fill probabilities, the evidence establishes apparent executable BBO triangles in historical data, not a deployable live strategy.

7 Conclusion

The cross-sectional results are consistent with liquidity and information-asymmetry theory. The less active GPRO book carries a much larger relative spread and signed price impact than AAPL.

However, the mechanism matters: GPRO's quoted spread is pinned by the tick, and raw share depth exaggerates its economic liquidity. AAPL's opening spread, impact, and variance move together, while its close remains quiet on this particular holiday-week session.

Foreign exchange follows a different clock. The London close produces the dominant spread, impact, and variance event, and EUR/JPY is persistently the widest and sparsest leg. Its negative one-minute autocorrelation indicates temporary reversal rather than a permanent common FX pattern. Triangular-arbitrage observations occur in the same broad afternoon window but are rare, short, and economically fragile.

The report therefore supports the standard theory only as far as the tape allows: greater relative illiquidity is associated with greater impact, but intraday shapes depend on tick constraints, trading intensity, market-specific clocks, and the finite identification available in one day of data. The Monday–Friday extension remains necessary to determine which shapes are persistent and which are specific to the selected sessions; the weekly implications stated in Section 2.2 should be treated as testable hypotheses rather than as additional findings.

A Function Timing Versus Opportunity Duration

Table 7: Wall-clock timing of the locked notebook run

Pipeline step	Seconds
Load and reconstruct NASDAQ books	136.022
Load and reconstruct currency books	316.155
Compute per-minute statistics	1.228
Construct price series, returns, and ACFs	0.648
Compute 30-minute diagnostics	0.291
Detect triangular arbitrage	0.529
Plot NASDAQ liquidity	8.221
Plot currency liquidity	9.479
Plot 30-minute variance and ACF	5.434
Plot full-sample ACF	3.392
Total timed wall clock	481.399

Notes: Timings are from the final notebook outputs and depend on hardware, local file access, and software state. Detection is timed after the synchronized one-second books already exist in memory.

The arbitrage scan takes 0.529 seconds once the books are built, compared with a mean episode length of 1.4 seconds and a maximum of 6 seconds. Thus, the vectorized historical detector is faster than the mean observed window. That comparison does not imply live feasibility. The 452 seconds spent loading and reconstructing historical books are an offline research cost, not the latency of an incremental feed handler. Conversely, a live system must receive three quotes, update three books, test both loops, route three aggressive orders, and obtain simultaneous fills before the opportunity closes. The present timing table measures only the Python research pipeline and cannot measure that end-to-end trading clock.

References

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[2] Albert S. Kyle. Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335, 1985.